Data Resources for Polygenic Risk Score (PRS) Model Development

To support the development of novel polygenic risk score (PRS) models, we are providing a comprehensive and diverse dataset, along with relevant tools and annotations. These resources are designed to facilitate robust training, evaluation, and interpretation of PRS models across multiple complex traits.

Training/Validation Dataset

- **Sample Size**: 8,663 individuals
    - 5,811 individuals in training
    - 2,852 individuals in validation
- **Base Location/Path:**
    - On HPC: **/gpfs/data/deciphEHRlab/deciphEHRdream/training_phase/**
    - On Databricks: **/Volumes/prod_ai_innovation/decipher_challenge/decipher_challenge_external/training_phase/**
- **Genetic data**:
    - In subdirectory: **genotype_files**
    - Genomic data is provided in both Variant Call Format (VCF) and PLINK binary format one file per chromosome from low coverage sequencing with imputation
        - Subdirectory: **vcf_files:** Contains vcf.gz and vcf.gz.tbi files with hard genotype calls from imputation, long with estimated nonreference allele dosage, and estimated genotype probabilities
        - Subdirectory: **plink_files**: Contains (.pgen/.pvar/.psam) binary files. Monomorphic variants are excluded and genotypes are set to missing when the alternate allele dosage is outside the range *[0, 0.1], [0.9, 1.1]*, or *[1.9, 2.0]*.
    - This facilitates ease of use across a variety of genetic analysis tools and pipelines.
- **Variant Information**:
    - Subdirectory: **supplementary_files**
    - Approximately **111 million variants** are included in this release.
    - Each variant is assigned a unique identifier to enable consistent referencing across supplementary files.

        - Subdirectory**: dbsnfp**
            - Functional prediction scores from dbNSFP v4.3a (a database from https://www.dbnsfp.org/ for scoring coding SNPs with various tools. Varity has been added but lifted over from hg37. Use with caution)
        - Subdirectory: **info_scores**
            - Measures of imputation quality per variant, facilitating filtering of poorly imputed variants.
            - Estimated info scores from imputation using the method presented in: https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-022-04863-z
        - Subdirectory: **snpeff**
            - Variant annotations for SnpEff version 5.1d. Information on annotations found here: https://pcingola.github.io/SnpEff/snpeff/inputoutput/
        - Subdirectory**: summary_statistics**
            - Selected GWAS summary statistics from the **GWAS Catalog** for the phenotypes listed below may be used to build PGS models. The corresponding files, along with documentation describing how they were generated and links to the relevant publications, can be found here: Link
- **Phenotypic and Demographic Data**:
    - Subdirectory: **phenotype_files**
        - Training_Cov_file.txt: contains covariate and disease status for the training dataset
        - Testing_Cov_file.txt: contains covariate information for the validation dataset
    - **Phenotypes:** Binary disease status is categorized by diagnosis groupers and is available for the following traits (0= no diagnosis, 1= diagnosis). The traits were defined using the Epic EHR diagnosis (EDG) concept, which is Epic's internal diagnosis/concept system used to group and normalize multiple diagnosis codes related to a given phenotype.
        - **Coronary artery disease (CAD)**
            - 2100000005: 'EDG CONCEPT CORONARY ARTERY DISEASE'
        - **Type II diabetes (T2D)**
            - 1765588: 'EDG BEST CARE TYPE 2 DIABETES POPULATION DEFINITION'
        - **Breast cancer (BCA)** (females only)
            - 1150000306: 'EDG CONCEPT CANCER BREAST'
    - Additional variables include:

- **Sex** (Male or Female)
- **Age** (capped at 89 and standardized)
- **Estimated genetic ancestry** (e.g., via principal components, standardized)